Module 08: Planning in Robotics

Learning Objectives

1. Define **Planning** within the context of Robotics.
2. Analyze how Planning interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of Planning.

Introduction

This module explores **Planning**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Planning is a critical component of the 8-part Active Inference spine, bridging the gap between Communication and Systems.

Key Concepts

1. Planning as a Markov Blanket Boundary

How does Planning define the boundary between the agent and the environment?

2. Generative Models of Planning

What parameters involved in Planning must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Planning drive the perception-action loop?

Applications

In Robotics, we see Planning manifest in: * **Specific Example 1**: A self-driving vehicle approaching an unprotected left turn plans by evaluating multiple policy trees over a 6-second horizon: each branch represents a different scenario for oncoming traffic behavior (yield, accelerate, change lane), and the planner selects the action sequence that minimizes expected free energy across all weighted scenarios -- this contingency planning naturally produces cautious behavior (waiting for a safe gap) when oncoming traffic intent is ambiguous, and decisive behavior (proceeding confidently) when the gap is clearly sufficient, without hand-coded rules for either case. * **Specific Example 2**: An autonomous exploration robot in an unknown environment plans its next movement by evaluating candidate viewpoints for their expected information gain using a frontier-based exploration strategy augmented with Active Inference; each candidate frontier cell is scored by the expected reduction in map entropy (epistemic value) discounted by the travel cost (pragmatic cost), and the planner selects the frontier that maximizes the ratio of information gained per meter traveled, producing efficient space-covering exploration trajectories that balance curiosity against energy expenditure.

Conclusion

Understanding Planning allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.